

ch1_working_with_java_data_types

ex1-

How many of these compile? 下列Comparator<String>的Lambda表達式中,有多少個能通過編譯?

- 18: Comparator<String> c1=(j,k)->0;
- 19: Comparator<String> c2=(String j,String k)->0;
- 20: Comparator<String> c3=(var j,String k)->0;
- 21: Comparator<String> c4=(var j,k)->0;
- 22: Comparator<String> c5=(var j,var k)->0;

D. 3

Comparator<String>的compare方法需要接收兩個String參數並回傳int,能夠匹配"兩個參數+一個整數結果"的寫法才可以編譯,參數個數不對,顯示參數型別與目標型別不兼容,都會編譯失敗

ans:D

ex2-

What is the output of the following application? 下面這應用程式序的輸出結果是甚麼?

```
public class Airplane{
    static int start=2;
    final int end;
    public Airplane(int x)
    {
        x=4;
        end=x;
    }

    public void fly(int distance)
    {
        System.out.print(end-start+" ");
        System.out.print(distance);
    }
}
```

```
public static void main(String... start)
{
    new Airplane(10).fly(5);

}

}
```

A. 2 5

建構子參數x先被重新賦值為4,因此final實例變數end的值是4,而不是創建對象時傳輸的10,靜態變數start為2, fly(5)先輸出end-start,也就是2,在輸出distance的值5

ans:A

ex3

Given the code fragment:

```
var i=10;
var j=5;
i+=(j*5+i)/j-2;
System.out.println(i);
```

What is the result? 給定代碼片段,運行結果是甚麼?

i初始為10,j為5,表達式 $i += (j * 5 + i) / j - 2$ 按整數運算計算: $(50 + 5) / 10 - 2 = 5 - 2 = 3$,因此i變為15

E) 15

ans:E

ex4

Given:

```
public class Tester{
    public static void main(String[] args){
        StringBuilder sb=new StringBuilder(5);
        sb.append("HOWDY");
        sb.insert(0, ' ');
        sb.replace(3,5,"LL");
        sb.insert(6,"COW");
        sb.delete(2,7);
        System.out.println(sb.length());
    }
}
```

What is the result? 給定StringBuilder操作代碼,最終輸出的長度式多少?

StringBuilder的append,insert,replace,delete都會直接修改同一個對象,需要按索引逐步修改字元串內容,delete(2,7)篩除的是從索引2(套件含)到7(不套件含)的字元~最終長度為4

D) 4

ans:D

ex5

Given:

```
public class StrBldr{
    static StringBuilder sb1=new StringBuilder("yo ");
    StringBuilder sb2=new StringBuilder("hi ");

    public static void main(String[] args){
        sb1=sb1.append(new StrBldr().foo(new StringBuilder("hey")));
        System.out.println(sb1);
    }
}
```

```
        StringBuilder foo(StringBuilder s){
            System.out.print(s+" oh "+sb2);
            return new StringBuilder("ey");
        }
    }
```

What is the result? 下面程序輸出會是甚麼?

方法呼叫參數new StringBuilder("hey")先傳入foo,foo輸出參數s加上實例變數sb2("hi"),隨後回傳新的StringBuilder("ey");main再把回傳值追加到靜態sb1("yo")後輸出,要注意StringBuilder是可變對象,但方法回傳的是另一個新對象

D) hey oh hi yo ey

ans:D

ex6

Given:

```
public class Test{
    public void process(byte v){
        System.out.println("Byte value "+v);
    }
    public void process(short v){
        System.out.println("Short value "+v);
    }
    public void process(Object v){
        System.out.println("Object value "+v);
    }
    public static void main(String[] args){
        byte x=12;
        short y=13;
        new Test().process(x+y);// line 1
    }
}
```

What is the output? 下面程序輸出會是甚麼?

A) Object value 25

x+y會先執行二次元數值提升;byte和short香家時都會提升為int,結果為int型別的25,多載方法沒有process(int),int可以裝為Integer,再向上轉型為object,所以呼叫process(object)

ans:A



ch2_controlling_program_flow

ex1

Variables declared as which of the following are never permitted in a switch statement? 聲明為下列哪兩種類型的變量,絕不能做為switch語句的選擇表達式(兩個選項)
(Choose two)

B. double

傳統switch支持byte,short,char,int及其套件裝型別,String和enum,不支持long,float,double,也不能直接使用普通object

F. Object

ans: B, F

ex2

What is the output of the following application? 下面應用程序的輸出結果是甚麼?

```
package planning;

public class ThePlan{
    var plan=1;
    plan=plan++ + --plan;
    if(plan==1){
        System.out.print("Plan A");
    }else{ if(plan==2) System.out.print("Plan B");
    }else System.out.print("Plan C");}
}
```

類別本體中直接寫了plan=plan+++++plan一類的可執行語句,但它不方法,建構子或是初始化程式碼中,所以語法不合法,即使表達方式本身可拆分,所以在位置上也會導致編譯失敗

D. The class does not compile

ans: D

ex3

Given:

```
int i=10;
do {
    for(int j=i/2;j>0;j--){
        System.out.print(j+" ");
    }
    i-=2;
}while(i>0);
```

What is the result? 給定嵌套循環代碼,運行結果是甚麼?

do while至少執行一次,每輪for迴圈從i/2開始遞減到1,隨後i在減2,因此打印
5~1,4~1,3~1,2~1,1

D) 5 4 3 2 1 4 3 2 1 3 2 1 2 1 1

ans:D

ch3_java_Object_Oriented_Approach

ex1

Which statement about the Elephant program is correct? 關於Elephant程序,下列何者說法正確?

```
package stampede;
interface Long{
    Number length();
}
```

```
public class Elephant{
    public class Trunk implements Long{
        public Number length(){ return 6;} //k1
```

```
}

public class MyTrunk extends Trunk{           //k2
    public Integer length(){ return 9;} //k3
}

public static void charge(){
    System.out.print(new MyTrunk().length());
}

public static void main(String[] cute){
    new Elephant().charge();                  //k4
}

}
```

介面方法回傳Number,Trunk用Number實作,MyTrunk覆寫時使用Integer,屬於協變回型別,是合法的,靜態charge方法也可以通過對象呼叫,所以程式碼本身可以編譯並輸出9

F. None of the above

ans:F

Ex2

What is the output of the following application? 下面應用程序的輸出結果是甚麼?

```
package sports;

abstract class Ball{
    protected final int size;
    public Ball(int size){
        this.size=size;
    }
}
```



```
interface Equipment{}
```

```
public class SoccerBall extends Ball implements Equipment{  
    public SoccerBall(){  
        super(5);  
    }  
    public Ball get(){ return this; }  
    public static void main(String[] passes){  
        var equipment=(Equipment)(Ball)new SoccerBall().get();  
        System.out.print(((SoccerBall)equipment).size);  
    }  
}
```

A. 5

SoccerBall建構子呼叫super(5),把父類別受保護的final欄位size初始化為5,對象先向上轉型為Ball,在轉型為Equipment,最後又向下轉回SoccerBall,實際對象始終是SoccerBall,因此讀取size得到5

Ans A

Ex3

Given:

```
public class GameObject{  
    public Object[] move(int x,int y){  
        System.out.println("Move GameObject");  
        return new Integer[]{x+10,y+10};  
    }  
}
```

and

```
public class Avatar extends GameObject{
    public Object[] move(Number x,Number y){
        System.out.println("Move Character");
        return super.move(x.intValue(),y.intValue());
    }

    public static void main(String... args)
    {
        var character=new Avatar();
        character.move(10.0,10.0);
        character.move(10,10);
    }
}
```

What is the result? [下面應用程序的輸出結果是甚麼？](#)

- A) Move Character
Move GameObject
Move GameObject

move(Number,Number)與父類別的move(int,int)參數不同,所以是多載而不是覆寫,傳入10.0時只能匹配Number版本,該方法內部呼叫父類別int版本,傳入整數10時,編輯器優先選擇更精確的int,int版本

ans:A

Ex4

```
public class Menu{
    enum Machine{
        AUTO("Truck"),MEDICAL("Scanner");
        private String type;
        private Machine(String type){
            this.type=type;
        }
        private void setType(String type){
            this.type=type;          //line 1
        }
        private String getType()
        {
            return type;
        }
    }
    public static void main(String[] args)
    {
        Machine.AUTO.setType("Sedan");    //line 2
        for(Machine p:Machine.values())
        {
            System.out.println(p+": "+p.getType()); //line 3
        }
    }
}
```

What is the result? 下面應用程序的輸出結果是甚麼?

列舉常量本質上是Machine的固定實例,但實例中的普通欄位仍然可以改變,AUTO的type被setType改成Sedan,MEDICAL保持Scanner

F) AUTO: Sedan
MEDICAL:Scanner

ans:F

ex5

Given:

```
class Scope{
    static int myint=666;
    public static void main(String[] args){
        int myint=myint;
        System.out.println(myint);
    }
}
```

Which is true? 關於下面變量作用代碼,哪項說法正確?

區域變數int myint=myint;中,右側名稱會被解析為正在宣告的區域變數,而不是同名靜態欄位,該區域變數尚未初始化,因此而出現"變數可牛尚未初始化"的編譯錯誤,可寫成Scope.myint

D) The code does not compile successfully

ans:D

Ex6

Given:

```
package test.t1;
public class A{
    public int x=42;
    protected A(){}    //line 1
}
```

and

```
package test.t2;
import test.t1.*;
```

```
public class B extends A{
    int x=17;           //line 2
    public B(){super();} //line 3
}
```

and

```
package test;
import test.t1.*;
import test.t2.*;
public class Tester{
    public static void main(String[] args){
        A obj=new B();           //line 4
        System.out.println(obj.x); //line 5
    }
}
```

What is the result? [關於下面變量作用代碼,哪項說法正確?](#)

B可以在建構子中呼叫父類別的A的protected建構子,欄位不具有執行時多太,obj的編譯時型別是A所以obj.x訪問的是A.x,輸出42,而不是B.x的17

G) 42

ans:G

Ex7

Given:

```
public class DNASynth{
```

```
int aCount;
int tCount;
int cCount;
int gCount;

DNASynth(int aCount,int tCount,int c,int g){
    // line 1
}
int setCCount(int c){
    return c;
}
void setGCount(int gCount){
    this.gCount=gCount;
}
}
```

Which two lines of code when inserted in line 1 correctly modifies instance variables? 在構造方法的line1插入哪兩行代碼,可以正確修改實力變量?(兩個選項)

B) `cCount=setCCount(c);`

E) `setGCount(g);`

ans:BE

`cCount=setCCount(c)`的方法回傳的c賦給實例欄位cCount,`setGCount(g)`通過this.gCount修改實例欄位,關鍵是區分參數名,區域變數名和對象欄位

Ex8

Given:

```
public class Price{
    private final double value;
    public Price(String value){
        this(Double.parseDouble(value));
    }
}
```

```
    }  
    public Price(double value){  
        this.value=value;  
    }  
    public Price(){}  
    public double getValue(){ return value;}  
    public static void main(String[] args)  
    {  
        Price p1=new Price("1.99");  
        Price p2=new Price(2.99);  
        Price p3=new Price();  
        System.out.println(p1.getValue()+" "+p2.getValue()+" "+p3.getValue());  
    }  
}
```

What is the result? [下面程序的運行結果是甚麼？](#)

final實例欄位value必須在每個構造路徑中被確定賦值,無參數建構子Price()沒有給value賦值,也沒有呼叫其他建構子,所以編譯器報錯

C) The compilation fails

ans:C

ex9

Given:

```
public interface Builder{  
    public A build(String str);  
}
```

and

```
public class BuilderImpl implements Builder{  
    @Override  
    public B build(String str){
```

```
        return new B(str);
    }
}
```

假設代碼能夠正確編譯,以下那三項說法正確?

Assuming that this code compiles correctly, which three statements are true?

實作介面方法時允許使用協變回傳型別,因此B必須是A的子類別型,方法本體中直接new B(str)
所以B不能是abstract,同時B要繼承A,因此A不能是final

D) B is a subtype of A.

E) B cannot be abstract.

F) A cannot be final.

ans:DEF

ex10

Given:

```
public class Foo{
    public void foo(Collection arg){
        System.out.println("Bonjour le monde");
    }
}
```

and

```
public class Bar extends Foo{
    public void foo(Collection arg){
        System.out.println("Hello world");
    }
    public void foo(List arg){
        System.out.println("Hello Mundol!");
    }
}
```



```
}
```

and

```
Foo f1=new Foo();  
Foo f2=new Bar();  
Bar b1=new Bar();  
List<String> li=new ArrayList<>();
```

Which three are correct? 基於給定的Foo,Bar和變量聲明,以下那三項說法正確?

多載在編譯時期根據參考型別和參數別選擇,覆寫在執行時期根據實際對象選擇,b1是Bar型別,List多載最具體,f2的編譯時型別是Foo,只能看到foo(Collection),執行實Bar的覆寫,f1則執行Foo版本

D) b1.foo(li) prints Hello Mundo!

E) f2.foo(li) prints Hello world

G) f1.foo(li) prints Bonjour le monde!

ans:DEG

ex11

Given:

```
public class Test{  
    public static void main(String[] args){  
        AnotherClass ac=new AnotherClass();  
        SomeClass sc=new AnotherClass();  
        ac=sc;  
        sc.methodA();  
        ac.methodA();  
    }  
}
```

```
class SomeClass{
    public void methodA(){
        System.out.println("SomeClass#methodA()");
    }
}
class AnotherClass extends SomeClass{
    public void methodA(){
        System.out.println("AnotherClass#methodA()");
    }
}
```

What is the Result? 下面代碼的運行果是甚麼？

ac的型別是AnotherClass,sc的型別是SomeClass,雖然sc實際參考AnotherClass對象,但不能不經強制型別轉換就把父類別型別參考賦予給子類別型別變數,因此ac=sc編譯失敗

D) The compilation fails

ans:D

Ex12

Given:

```
interface AbilityA{
    default void action(){
        System.out.println("a action");
    }
}
```

and

```
interface AbilityB{
    void action();
}
```

and

```
public class Test implements AbilityA,AbilityB{ // line 1
    public void action() {
        System.out.println("ab action");
    }
    public static void main(S[] args){
        AbilityB x=new Test();           //line 2
        x.action();
    }
}
```

What is the result? 下面代碼的運行結果是甚麼？

AbilityA提供default action,AbilityB宣告抽象action,實作類別Test自己覆寫action,解決了繼承衝突,通過AbilityB參考呼叫時仍會發生動態綁定,執行Test.action

E) ab action

ans:E

Ex13

Given the enum declaration:

```
1.enum Alphabet{  
2.    A,B,C  
3.  
4.}
```

Example this code:

```
System.out.println(Alphabet.getFirstLetter());
```

What code should be written at line 3 to make this code print A? 在枚舉地3行寫入哪段代碼才能使示例輸出A?

A) static String getFirstLetter() { return A.toString();}

定義靜態方法getFirstLetter()並返回A.toString()

列舉常量可以通過類別名稱直接訪問,靜態方法回傳第一個列舉常量A的字元串刑事,因此println輸出A
ans:A

ex14

Given:

```
public interface ExampleInterface{
```

Which two statements are valid to be written in this interface? 在空接口ExampleInterface中,哪兩個方法聲明是合法的?

A) public String method();

介面中的普通抽象方法即使不顯式寫public abstract,也會隱式具有這兩個修飾符,不能使用與介面衝突的訪問權限或非方法本體

G) public abstract void methodB();

ans:AG

Ex15

Given:

```
public class Person{
    private String name;
    private Person child;
    public Person(String name,Person child)
    {
        this(name);
        this.child=child;
    }
    public Person(String name){
        this.name=name;
    }
    public String toString(){
        return name+" "+child;
    }
}
```

and

```
public class Tester{
    public static Person createPeople(){
        Person jane=new Person("Jane");
        Person john=new Person("John",jane);
        return jane;
    }
}
```

```
public static Person createPerson(Person person){
    person=new Person("Jack",person);
    return person;
}
public static void main(String[] args)
{
    Person person=createPeople();
    // line 1
    person=createPerson(person);
    // line 2
    String name=person.toString();
    System.out.println(name);
}
}
```

Which statement is true? [關於對象引用與垃圾回收哪項說法正確?](#)

createPeople回傳jane,john只是區域變數,方法結束後不在可被參考到,createPerson創建Jack,並讓Jack.child指向原來的Jane,當line2執行後,person指向Jack,Jack仍被Jack.child參考,所以嚴格說此時Jane仍可以被參考到

C) The memory allocated for Jane object can be reused in line 2.
在line2,Jane對象占用的內存可以被回收並重新利用

ans:C

ex16

Given:

```
public interface A{
    public Iterable a();
}
```

```
public interface B extends A{
    public Collection a();
}
```

```
public interface C extends A{
    public Path a();
}
public interface D extends B,C{

}
```

why does D cause a compilation error? 為什麼接口D會導致編譯錯誤?

D同時從B和C繼承a(),但返回類型不兼容

C) D inherits a() from B and c but the return types are incompatible.

B.a()回傳Collection,C.a()回傳Path,兩者都可以作為A.a()回傳Iterable的協作覆寫,但Collection與Path之間沒有子類別型關係,D無法繼承出一個唯一兼容的方法

ans:C

Ex17

Given the code fragment:

```
8. public class Test {
9.     private final int x=1;
10.    static final int y;
11.    public Test() {
12.        System.out.print(x);
13.        System.out.print(y);
14.    }
15.    public static void main(String args[]) {
16.        new Test();
17.    }
18.}
```

what is the result? 下面代碼的運行結果是甚麼?

A) The compilation falls at line 13

Static final欄位y沒有在宣告處或靜態初始化中賦值,建構子中讀取y時,編譯器發現他尚未完成確定賦值,因此無法編譯

ans: A

ex18

Given:

1. interface Pastry {
2. void getIngredients();
3. }
4. abstract class Cookie implements Pastry{
- 5.
6. class ChocolateCookie implements Cookie {
7. public void getIngredients(){}
8. }
9. class CoconutChocolateCookie extends ChocolateCookie {
10. void getIngredients(int x){}
11. }

Which is true? 關於下面Pastry, Cookie和ChocolateCookie的代碼哪項正確?

Cookie式類別不是介面,因此ChocolateCookie應該使用extendsCookie,而不能implements Cookie,抽象類別可以暫時不時做介面方法,但具體子類別必須實作

D) The compilation falls due to an error in line 6. 第六行導致編譯失敗

ans:D

ch4_Exception_Handling

ex1

Given the following application, which specific type of exception will be printed in the stack trace at runtime? 運行時,堆線跟蹤最終打印的是哪種具體異常?

```
package carnival;

public class WhackAnException{
    public static void main(String... hammer)
    {
        try{
            throw new ClassCastException();
        }catch(IllegalArgumentException e){
            throw new IllegalArgumentException();
        }catch(RuntimeException e){
            throw new NullPointerException();
        }finally{
            throw new RuntimeException();
        }
    }
}
```

try中拋出的ClassCastException會被RuntimeException的catch捕捉並改拋NullPointerException,但finally無論如何都會執行,並再次拋出RuntimeException,finally拋出的例外覆蓋之前正在傳播的例外

D. RuntimeException

ans:D

Ex2

下面再try-with-resources程序在e1行打印的主異常類名是甚麼？

Given the following application, what is the name of the class printed at line e1?

```
package canyon;

final class FallenException extends Exception{}

final class HikingGer implements AutoCloseable{
    @Override public void close() throws Exception{
        throw new FallenException();
    }
}

public class Cliff{
    public final void climb() throws Exception{
        try(HikingGear gear=new HikingGear()){
            throw new RuntimeException();
        }
    }
    public static void main(String... rocks){
        try{
            new Cliff().climb();
        }catch(Throwable t){
            System.out.println(t); //e1
        }
    }
}
```

B. java.lang.RuntimeException

try主體先拋RuntimeException,隨後資源close()又拋FallenException,try-with-resources會保留主體例外做為主例外,把關閉資源產生的例外放入suppressed exceptions,所以接打印t時顯示 RuntimeException

ans:B

Ex3

Given:

```
import java.io.FileNotFoundException;  
import java.io.IOException;  
  
public class Tester{  
    public static void main(String[] args){  
        try{  
            doA();  
        }//line 1  
    }  
    private static void doA() throws Exception,IndexOutOfBoundsException{  
        if(false){  
            throw new FileNotFoundException();  
        }else{  
            throw now IndexOutOfBoundsException();  
        }  
    }  
}
```

What must be added in line 1 to compile this class? 為了使該類通過編譯,line1必須添加甚麼?

C) catch(Exception e){}

doA()宣告throws Exception,因此呼叫方法必須捕捉Exception或繼續宣告拋出,IndexOutOfBoundsException是非受檢例外,不強制處理,但Exception是受檢例外

ans:C

Ex4

Given:

```
char[] characters=new char[100];
try(FileReader reader=new FileReader("file_to_path")){
    // line 1
    System.out.println(String.valueOf(characters));
} catch(IOException e){
    e.printStackTrace();
}
```

You want to read data through the reader object.

Which statement inserted on line 1 will accomplish this? 要通過reader對象讀取數據,在line1應插入哪條語句?

FileReader.read(char[])會嘗試把字元毒入給定陣列,並回傳實際讀取數量或-1,try-with-resources會在結束時自動關閉reader

C) reader.read(characters);

ans:C

ex5

Given:

```
public class ExSuper extends Exception{
    private final int eCode;
    public ExSuper(int eCode,Throwable cause){
        super(cause);
        this.eCode=eCode;
    }

    public ExSuper(int eCode,String msg,Throwable cause){
        super(msg,cause);
        this.eCode=eCode;
    }
}
```

```
    public String getMessage(){
        return this.eCode+":
"+super.getMessage()+"_"+this.getCause().getMessage();
    }
}
```

```
public class ExSub extends ExSuper{
    public ExSub(int eCode,String msg,Throwable cause)
    {
        super(eCode,msg,cause);
    }
}
```

and the code fragment:

```
try{
    String param1="oracle";
    if(param1.equalsIgnoreCase("oracle")) {
        throw new ExSub(9001, "APPLICATION ERROR-9001",new
FileNotFoundException("MyFile.txt"));
    }
    throw new ExSuper(9001, new FileNotFoundException("MyFile.txt")); //
Line 1
}
catch(ExSuper ex)
{
    System.out.println(ex.getMessage());
}
```

What is the result? [下面自定義異常程序的運行結果是甚麼？](#)

參數等於oracle,所以拋出ExSub,ExSub呼叫父類別含消息和cause的建構子,覆寫的getMessage()把錯誤碼,父類別消息和cause的消息拼接起來

C) 9001: APPLICATION ERROR-9001-MyFile.txt

ans: C

Ex6

Given:

```
public class Option{
    public static void main(String[] args) {
        System.out.println("Ans: "+convert("a").get());
    }

    private static Optional<Integer> convert(String s) {
        try{
            return Optional.of(Integer.parseInt(s));
        } catch (Exception e) {
            return Optional.empty();
        }
    }
}
```

What is the result? [下面Optional程序的運行結果是甚麼?](#)

C) A java.util.NoSuchFilementException is thrown at run time

ans:C

Ex7

Given:

```
import java.io.*;
public class Tester {
    public static void main(String args[])
    {
        try{
```

```
        doA();
        doB();
    } catch(IOException e) {
        System.out.print("C");
        return;
    } finally {
        System.out.print("d");
    }
    System.out.print("f");
}
private static void doA() {
    System.out.print("a");
    if(false) {
        throw new IndexOutOfBoundsException();
    }
}
private static void doB() throws FileNotFoundException {
    System.out.print("b");
    if(true) {
        throw new FileNotFoundException();
    }
}
}
```

What is the result? 下面程序的運行結果是甚麼?

D) abcd

doA輸出a;doB先輸出b再拋FileNotFoundException,catch(IOException)輸出c並執行return,但return之前finally仍執行衝d;finally後面的f在執行

ans:D

Ex8

Given:

```
public class Test{
    private int num=1;
    private int div=0;

    public void divide() {
        try {
            num=num/div;
            System.out.print("Exception");
        }
        catch(ArithmeticException ae) { num=100; }
        catch(Exception e) { num=200; }
        finally { num=300; }
        System.out.print(num);
    }
    public static void main(String args[])
    {
        Test test=new Test();
        test.divide();
    }
}
```

What is the output? 下面divide()運行結果是甚麼?

C) 300

1/0拋ArithmeticException,進入第一個catch把num設為100,隨號finally必定執行,把num改為300,最後打印300

ans: c

Ex9

Given:

```
public class Test {  
    private int sum;  
    public int compute() {  
        int x=0;  
        while(x<3) {  
            sum+=++x;  
        }  
        return sum/4;  
    }  
  
    public static void main(String[] args) {  
        Test t=new Test();  
        int sum=t.compute();  
        sum=t.compute();  
        System.out.print(sum);  
    }  
}
```

What is the output? 下面程序的輸出結果是甚麼?

B) 3

第一次compute把實例欄位sum從0累加1+2+3=6,回傳整數除法6/4=1,第二次繼續在同一個欄位上累加到12,回傳12/4=3,最後打印第二次回傳值3

ans:B

ch5_Working_with_Arrays_and_Collections

ex1

Which of the following fills in the blank so this code compiles? 在通配符空白處填入甚麼代碼才能編譯?

```
public static void getExceptions(Collection<__> coll){  
    coll.add(new RuntimeException());  
    coll.add(new Exception());  
}
```

方法需要向集合中加入Exception和RuntimeException,使用Collection<?super Exception>表示元素型別是Exception或其父類別,因此這兩個對象都可以安全加入

C. ? super Exception

ans:C

Ex2

What does the following output? 下面集合代碼會產生甚麼結果?

```
18: List<String> list=List.of(  
19:     "Mary","had","a","little","lamb");  
20: Set<String> set=new HashSet<>(list);  
21: set.addAll(list);  
22: for(String sheep:set)  
23:     if(sheep.length()>1)  
24:         set.remove(sheep);  
25: System.out.println(set);
```

增強for迴圈底層使用Iterator,編歷HashSet時直接呼叫set.remove(sheep)屬於結構性修改,迭代器檢測到集合被併行修改後會拋出ConcurrentModificationException

E. The code throws an exception at runtime 運行時出現異常

ans:E.

ex3

Given:

```
ArrayList<Integer> a1=new ArrayList<>();
a1.add(1);
a1.add(2);
a1.add(3);
Iterator<Integer> itr=a1.iterator();
while(itr.hasNext()) {
    if(itr.next()==2) {
        a1.remove(2);
        System.out.print(itr.next());
    }
}
```

What is the result? 下面代碼會產生甚麼結果?

C) A ConcurrentModificationException is thrown at run time

ans:C

Ex4

Given:

```
import java.util.ArrayList;
import java.util.Arrays;
public class NewMain{
    public static void main(String[] args) {
        String[] catNames={"abyssinian","oxicat",
            "korat","laperm","bengal","sphynx"};
```

```
var cats=new ArrayList<>{Arrays.asList(catNames));  
cats.sort((var a,var b)->-a.compareTo(b));  
cats.forEach(System.out::println);  
}  
}
```

What is the result?

B) sphynx

oxicat

laperm

korat

bengal

abyssinian



ans:B

ch6_Working_with_Streams_and_lambda

ex1

What is a possible output of the following application?

```
import java.util.*;
import java.util.stream.*;

public class Thermometer {
    private String feelsLike;
    private double temp;
    @Override public String toString(){ return feelsLike;}
    // Constructor/Getters/Setters Omitted

    public static void main(String... season) {
        var readings=List.of(new Thermometer("HOT!",72),
            new Thermometer("Too Cold",0),
            new Thermometer("Just right!"72));
        readings
            .parallelStream()           // k1
            .collect(Collectors.groupingByConcurrent(
                Thermometer::getTemp)) // k2
            .forEach(System.out::println); // k3
    }
}
```

E) The code does not compile because of line k3

ans:E

ex2

What is the output of the following application?

```
package lot;

import java.util.function.*;

public class Warehouse {
    private int quantity=40;
    private final BooleanSupplier stock;
    {
        stock=()->quantity>0;
    }

    public void checkInventory() {
        if(stock.get())
            System.out.print("Plenty!");
        else {
            System.out.print("On Backorder!");
        }
    }

    public static void main(String... widget) {
        final Warehouse w13=new Warehouse();
        w13.checkInventory();
    }
}
```

C. The code does not compile because of the checkInventory() method.

ans:C

ex3

Which code fragment represents a valid Comparator implementation?

```
D) new Comparator<String>() {  
    public int compare(String str1,String str2) {  
        return str1.compareTo(str2);  
    }  
};
```

ans:D

Ex4

Given:

```
var fruits=List.of("apple","orange","banana","lemon");  
Optional<String> result=fruits.stream().filter(f->f.contains("n")).findAny();// line 1  
  
System.out.println(result.get());
```

You replace the code on line 1 to use parallelStream.

Which one is correct?

D) The code may produce a different result

ans:D

ex5

Given the code fragment:

1. var list=List.of(1,2,3,4,5,6,7,8,9,10);
2. UnaryOperator<Integer> u=i->i*2;
3. list.replaceAll(u);

Which can replace line 2?

C) UnaryOperator<Integer> u=(var i)->(i*2);

ans:C

Ex6

Which two are valid statements?

B) BiPredicate<Integer,Integer> test=(Integer x,final Integer y)->(x.equals(y));

D) BiPredicate<Integer,Integer> test=(var x,final var y)->(x.equals(y));

ans:BD

Ex7

Why would choose to use a peek operation instead of a forEach operation on a Stream?

A) to process the current item and return a stream

ans:A

Ex8

Given the content from lines.txt

C

C++

Java

Go

Kotlin

and

```
String fileName="lines.txt";  
List<String> list=new ArrayList<>();  
try(Stream<String> stream=Files.lines(Paths.get(fileName))) {  
    list=stream  
        .filter(line->!line.equalsIgnoreCase("JAVA"))  
        .map(String::toUpperCase)  
        .collect(Collectors.toList());  
} catch(IOException e) {  
}  
list.forEach(System.out::println);
```

What is the Result?

C) C

C++

GO

KOTLIN

ans:C

Ex9

Given:

```
public class Employee {  
    private String name;  
    private String neighborhood;  
    private int salary;  
    //Constructors and setter and getter methods go here  
}
```

and the code fragment:

```
List<Employee> roster=new ArrayList<>();  
Predicate<Employee> p=e->e.getSalary()>30;  
Function<Employee,Optional<String>> f=  
    e->Optional.ofNullable(e.getNeighborhood());
```

Which two objects group all employees with a salary greater than 30 by neighborhood?

A) `Map<Optional<String>,List<Employee>> r4=roster.stream()
 .collect(Collectors.groupingBy(f,Collectors.filtering(p,Collectors.toList())));`

```
E) Map<String,List<Employee>> r1=roster.stream()  
    .collect(Collectors.groupingBy(Employee::getNeighborhood,  
    Collectors.filtering(p,Collectors.toList())));
```

ans:AE

Ex10

Given the code fragment:

```
public class Main {  
    public static void main(String[] args)  
    {  
        List<String> fruits=List.of("banana","orange","apple","lemon");  
        Stream<String> s1=fruits.stream();  
        Stream<String> s2=s1.peek(i->System.out.print(i+" "));  
        System.out.println("-----");  
        Stream<String> s3=s2.sorted();  
        Stream<String> s4=s3.peek(i->System.out.print(i+" "));  
        System.out.println("-----");  
        String strFruits=s4.collect(Collectors.joining(","));  
    }  
}
```

What is the output?

A) -----

banana orange apple lemon apple banana lemon orange

ans:A



ch7_Java_Platform_Module_System

ex1

What statement are true about requires mandated java.base?(Choose two)

C. This output is expected when running the jdeps command

E. All modules will include this in the output

ans:CE

ex2

What is the name of a file that declares a module?

F. module-info.java

ans:F

ex3

Suppose you have a module that contains a class with a call to `exports(ChocolateLab.class)` . Which part of the module service contains this class?

E. None of the above

ans:E

ex4

How many of these keywords can be used in a module-info.java file:close,export,import,require,and uses?

B. One

ans:B

exports,requires,transitive,provides,opens,uses

ex5

Which module defaults the foundational APIs of the Java SE Platform?

B) java.base

ans:B

ch8_Concurrency

ex1

What is the output of the following application?

```
import java.util.*;
```

```
public class SearchList<T> {  
    private List<T> data;  
    private boolean foundMatch=false;  
    public SearchList(List<T> list) {
```

```
        this.data=list;
    }
    public void exists(T v,int start,int end) {
        if(end-start==0) {}
        else if(end-start==1) {
            foundMatch=foundMatch || v.equals(data.get(start));
        } else {
            final int middle=start+(end-start)/2;
            new Thread(()->exists(v,start,middle)).run();
            new Thread(()->exists(v,middle,end)).run();
        }
    }

}

public static void main(String[] a) throws Exception {
    List<Integer> data=List.of(1,2,3,4,5,6);
    SearchList<Integer> t=new SearchList<Integer>(data);
    t.exists(5,0,data.size());
    System.out.print(t.foundMatch);
}

}
```

A. true

ans:A

Ex2

Which of the following methods is not available on an `ExecutorService` instance?
(Choose two.)

A. `execute(Callable)`

D. `exit()`

ans:A,D

ex3

Given:

```
var c=new CopyOnWriteArrayList<>(List.of("1","2","3","4"));
```

```
Runnable r={()->{
```

```
    try{
```

```
        Thread.sleep(150);
```

```
    }
```

```
    catch(InterruptedException e){
```

```
        System.out.println(e);
```

```
    }
```

```
    c.set(3,"four");
```

```
    System.out.print(c+" ");
```

```
};
```

```
Thread t=new Thread(r);
```

```
t.start();
```

```
for(var s:c)
```

```
{
```

```
    System.out.print(s+" ");
```

```
    Thread.sleep(100);
```

```
}
```

What is the output?

A) 1 2 [1, 2, 3, four] 3 4

ans:A

ex4

Given:

```
public interface Worker {  
    public void doProcess();  
}
```

and

```
public class HardWorker implements Worker {  
    public void doProcess() {  
        System.out.println("doing things");  
    }  
}
```

and

```
public class Cheater implements Worker {  
    public void doProcess(){}  
  
}
```

and

```
public class Main<T extends Worker> extends Thread { // Line 1
    private List<T> processes=new ArrayList<>(); //line 2
    public void addProcess(HardWorker w) { // line 3
        processes.add(w);
    }

    public void run() {
        processes.forEach((p)->p.doProcess());
    }
}
```

What needs to change to make these classes compile and still handle all types of Interface Worker?

B) Replace Line 3 with public void addProcess(T w) {

ans:B

ch9_Java_IO_API

ex1

Why does Console readPassword() return a char array rather than a String?

B. It improves security

ans:B

ex2

Fill in the blanks:Writer is a(n)_____ that related stream classes_____

B. abstract class, extend

ans:B

ex3

Given:

```
class MyPersistenceData {  
    String str;  
    private void methodA() {  
        System.out.println("methodA");  
    }  
}
```

You want to implement the java.io.Serializable interface to the MyPersistenceData class.

Which method should be overridden?

D) nothing

ans:D

ch10_Secure_Coding_in_java_se_Application

ex1

Fill in the blanks: _____ means the state of an object cannot be changed, while _____ means that it can

B. Immutability, mutability //不變性，可變性

ans:B

ex2

How do you change the value of an instance variable in an immutable class?

E. You can't

ans:E

定義 immutable class 的方法如下

不提供修改屬性內容的 **setter** 方法。

所有的屬性宣告為 **private final** 使其無法被修改。

類別宣告為 **final class** 使無法被繼承，方法無法被覆寫。或是把建構式設為 **private**，並改以工廠方法提供物件的實例。

若屬性為 **mutable object** 應避免提供修改方法，讀取時提供複製物件而非原物件的參照。

Ex3

Given:

```
public class Foo {  
    public static String ALPHA="alpha";  
    protected String beta="beta";  
    private final String delta;  
    public Foo(String d) {  
        delta=ALPHA+d;  
    }  
    public String foo() {  
        return beta+=delta  
    }  
}
```

Which change would make Foo more secure?

D) public static final String ALPHA="alpha";

ans:D

ch11_Database_Application_with_JDBC

ex1

What is the most likely outcome of this code if the bunnies table is empty?

```
var url="jdbc:derby:bunnies";  
var sql="insert into bunny(name,color) values(?,?)";  
try(var conn=DriverManager.getConnection(url);  
var stmt=conn.createStatement()){  
    stmt.setString(1,"Hoppy");  
    stmt.setString(2,"Brown");  
}
```

```
        stmt.executeUpdate(sql);  
    }
```

C. The code does not compile.

ans:C

Ex2

What must be the first characters of a database URL?

D. jdbc:

ans:D

ex3

Assuming the user credentials are correct, which expression will create a Connection?

A) DriverManager.getConnection("jdbc:derby:com");

ans:A

ch12_Localization

ex1

Which of the following are considered locales?(Choose three)

A. Culture region

E. Political region

F. Geographical region

ans:AEF

ex2

When localizing an application, which type of data varies in presentation depending on locale?

C. Both

ans:C

ex3

Given the code fragment:

```
Locale locale=Locale.US;
```

```
// Line 1
```

```
double currency=1_00.00;
```



```
System.out.println(formatter.format(currency));
```

You want to display the value of currency as \$100.00.

Which code inserted on line 1 will accomplish this?

```
1 D) NumberFormat formatter=NumberFormat.getCurrencyInstance(locale);
```

ans:D

ex4

Which code fragment does a service use to load the service provider with a Print interface?

```
A) private java.util.ServiceLoader<Print> loader=ServiceLoader.load(Print.class);
```

ans:A

ch13_Annotations

ex1

What modify is used to mark an annotation element as optional?

B. default

ans:B

ex2

Fill in the blank with the correct annotation usage that allows the code to compile without any warnings.

```
@Deprecated(since="5.0")
public class ProjectPlanner<T>{
    ProjectPlanner create(T t) { return this; }
}
```

```
@SuppressWarnings(_____)
class SystemPlanner {
    ProjectPlanner planner=new ProjectPlanner().create("TPS");
}
```

D. {"deprecation","unchecked"}

ans:D

ex3

Given the declaration:

```
@interface Resource {  
    String[] value();  
}
```

Examine this code fragment:

```
/** Loc1**/* class ProcessOrders{...}
```

Which two annotation may be applied at Loc1 in the code fragment?

D) @Resource({"Customer1","Customer2"})

E) @Resource("Customer1")

ans:DE

